

Children Listening to Music or Watching Cartoons During ER Procedures: A RCT

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Abstract

Objective This study aims to determine if listening to music and watching cartoons are effective to distract children from pain and distress during procedures in the emergency room (ER).

Methods This study is a single-center, 3-armed, superiority randomized controlled trial comparing listening to music, watching cartoons, and standard care during ER procedures in children aged 3–13 years. The primary outcome was pain measured from video footage with the Alder Hey Triage Pain Score (AHTPS). Children older than 4 years self-reported pain with the Faces Pain Scale-Revised (FPS-R). The secondary outcome was distress measured with the Observational Scale of Behavioral Distress-revised (OSBD-r). Another indicator of distress was heart rate.

Results Data of 191 participants were analyzed for the 3 groups: music ($n = 75$), cartoon ($n = 62$), and control ($n = 54$). The median age was 7.3 years (4.9–9.7). In multivariable analysis, pain assessed with the AHTPS was significantly lower ($B = -1.173$, 95% confidence interval -1.953 , -0.394 , $p = .003$) in the music group than in the control groups. Across the 3 groups, 108 children self-reported pain with the FPS-R after the procedure. The scores were lowest in the music group, but the differences between groups were not significant ($p = .077$). OSBD-r distress scores assigned during the procedures were not significantly different between the 3 groups ($p = .55$). Heart rate directly after the procedure was not statistically significantly different between the 3 groups ($p = .83$).

Conclusions Listening to recorded music is a beneficial distraction for children experiencing pain during ER procedures, whereas watching cartoons did not seem to reduce pain or distress.

Key words: cartoons; children; distraction; emergency room; music; procedural pain.

Introduction

Having to undergo a medical procedure in the emergency room (ER) can be very upsetting for children (Blount, Piira, Cohen, & Cheng, 2006). The resulting pain and distress should be minimized to avoid development of fear for medical procedures (American Academy of Pediatrics, Committee on Psychosocial Aspects of Child and Family Health; Task Force on

Pain in Infants, Children, and Adolescents, 2001; Kennedy, Luhmann, & Zempsky, 2008; Leahy et al. 2008; Rocha, Marche, & von Baeyer, 2009). Current pain management for children in the ER can include, in addition to medication, nonpharmacological approaches such as touch, positioning, application of heat, or cold compresses and distraction (Krauss,

Calligaris, Green, & Barbi, 2016). Depending on the level of interaction required by the patient, a range of active and passive distraction techniques can be used (MacLaren & Cohen, 2005).

Distraction is a shifting of attention away from distress, pain or painful stimuli to stimuli that are more engaging or enjoyable (Birnie, Chambers, & Spellman, 2017). Active distraction techniques include blowing bubbles and balloon inflation (Longobardi, Prino, Fabris, & Settanni, 2019; Mutlu & Balci, 2015; Sahiner & Bal, 2016), hypnosis (Birnie et al., 2014), vibration (Moadad, Kozman, Shahine, Ohanian, & Badr, 2016), playing with toys (Hillgrove-Stuart, Pillai Riddell, Horton, & Greenberg, 2013; MacLaren & Cohen, 2005), and individual expressive music therapy (Bradt, Dileo, & Shim, 2013; Bradt et al., 2015; Caprilli, Anastasi, Grotto, Scollo Abeti, & Messeri, 2007; Gold et al., 2011; Hartling et al., 2013; Klassen, Liang, Tjosvold, Klassen, & Hartling, 2008; Noguchi, 2006). Passive distraction techniques include amongst others listening to recorded music or watching a movie (MacLaren & Cohen, 2005; Nilsson, Enskar, Hallqvist, & Kokinsky, 2013).

The mechanism of passive distraction methods is still unclear (Bagnasco, Pezzi, Rosa, Fornonil, & Sasso, 2012; Bergomi, Scudeller, Pintaldi, & Dal Molin, 2018). Possible processes of distraction are attributed to suppressing the nervous system activities associated with pain and diverting attention away from noxious stimuli (Felluga et al., 2016; Johnson, 2005; Sahiner & Bal, 2015). Although there is conflicting data about whether active or passive distraction is more effective (Nilsson et al., 2013), recent systematic reviews show strong support for both types of distraction in reducing needle-related pain and distress in children (Birnie et al., 2014; Birnie, Noel, Chambers, Uman, & Parker, 2018). Four systematic reviews support possible beneficial effects of distraction in reducing distress and pain during painful procedures in the ER (Birnie et al., 2014; Birnie et al., 2018; Uman et al., 2013; Wente, 2013). The evidence, however, comes from articles assessed to be of low quality. Furthermore, the question which distraction intervention is best to alleviate children's pain and distress during the wide range of procedures in the ER remains unsolved. Not all interventions are easy to implement and some require extra staff. Costly distraction interventions may not always be feasible in resource-limited settings, but passive distraction interventions such as listening to music and watching cartoons do not require many extra resources and are relatively easy to implement.

Although both music and cartoon interventions seem beneficial in reducing children's distress and pain in during ER procedures, the evidence is still inconclusive (Cohen, Blount, & Panopoulos, 1997; Downey & Zun, 2012; Hartling et al., 2013). A systematic review of Klassen et al. (2008) suggests that listening to

recorded music during painful procedures might have more benefits than live music therapy.

Bellieni et al. (2006) found that watching cartoons was more effective in lowering children's self-reported pain scores than was active distraction and a control condition. Two other studies found that watching cartoons was effective in reducing children's distress and pain during immunizations and in an ER setting during painful procedures (Cohen et al., 1997; Downey & Zun, 2012). Furthermore, most studies have been performed in the US and Canada. Testing these interventions in resource-limited ERs seems relevant as admissions numbers are typically much higher, while distress levels are bound to be similar.

Our hypothesis is that listening to music or watching cartoons will be more beneficial than standard care alone. Because of the lack of studies in the literature, we cannot hypothesize which of the two distraction techniques will be more effective. Therefore, the objective of this randomized controlled trial (RCT) was to explore if listening to recorded music or watching cartoons can reduce pain and distress in children aged 3–13 years during medical procedures performed in the ER in a South African setting.

Materials and Methods

Study Design

This single-center, three-armed, superiority RCT was conducted from March 2014 through September 2014 at two sites: the surgical trauma unit and medical ER of the Red Cross War Memorial Children's Hospital (RCWMCH), Cape Town, South Africa, a tertiary pediatric trauma center that treats approximately 6500 children up to 13 years old.

Setting and Participants

All children aged 3 up to 13 years attending the surgical trauma unit and medical ER were eligible if they were undergoing: venepuncture; IV placement; application of a temporary splint or plaster cast; injection of local anesthetics; wound dressing; and suturing of open wounds. If more than one procedure was performed, only the first procedure was included. The exclusion criteria were the following: hearing impairment, developmental disability, or no responsiveness to stimuli from the environment due to altered level of consciousness. In addition, children were excluded for whom the attending staff felt that the study would interfere with the treatment or if seeking consent would be too delicate, for example, in case of suspected intentional trauma.

Interventions

The music and cartoon interventions were carried out by the researchers (H. Mevius and N. van der Heijde).

The music intervention consisted of listening to recorded music of a repetitive and relaxing quality. To avoid possible negative connotation with familiar music, we asked a music therapist to compose several original songs with a variety of rhythms, instruments, and lyrics. The recorded music was played through ambient speakers, set at a fixed volume, connected to an iPod dock on repeat for the duration of the procedure.

The cartoon intervention consisted of watching Disney's "Chip and Dale" cartoon animations, selected by a pediatric psychologist, a total of three consecutive clips each of 10 min were available. "Chip and Dale" cartoons are simple, funny stories about the adventures of two chipmunks. The cartoon animations were shown on a laptop connected to speaker boxes, set at a fixed volume, for the duration of the procedure. The laptop was placed on a table or held by the researcher such that the child could comfortably watch the movie. The researcher made sure the laptop was visible for the child, whether the child was sitting or lying down. The control group received standard care, which could include comforting words from attending family or medical staff as is normal in ER practice.

Study Procedure

Prior to starting the RCT, two members of the research team were trained to recruit and enroll patients. After a child's admission to the trauma unit, one of the researchers would explain the study to the accompanying parent or caregiver and seek informed consent. Children aged 7 years or older needed to provide assent. Upon inclusion, the child was randomly allocated to one of the three study arms: music intervention, cartoon intervention, or the control group. If the child did not want to continue with the intervention, the music or cartoon would be turned off, but the child would still be included in an intention-to-treat analysis. Parents were encouraged to interact with their child as they saw fit in all three groups.

We videotaped the procedure of the medical treatment and the intervention or control condition using a video camera on a tripod aiming to focus on the child's face and body. The interventions were administered from the start of the treatment. Start of the Plaster of Paris (POP) back-slab procedure was defined as the placement of the first cotton padding; the end of the treatment was defined as placing tape on the cast. Start of the other treatment procedures was defined as cleaning the injection or wound site and the end as placing tape or after the last injection.

Outcome Measures

The primary outcome was pain during an ER procedure, as measured with the Alder Hey Triage Pain

Score (AHTPS), which was specifically designed to measure pain during painful procedures. This scale has been validated for all age groups in the ER setting in U.K. (Stewart, Lancaster, Lawson, Williams, & Daly, 2004) and in the ER of the RCWMCH (Thiadens, Vervat, Albertyn, Van Dijk, & Van As, 2011). The scale assesses five behaviors: cry, facial expression, posture, movement, and pallor. All items are scored from 0 to 2 with a total score between 0 and 10. The AHTPS was scored once to represent the entire medical treatment. Following the AHTPS guidelines, the assessors were instructed to score the most distressed display of pain behavior in one session. Furthermore, children older than 4 years self-reported pain before and after the medical procedure with the Faces Pain Scale-Revised (FPS-R), which has been validated for children from 4 to 16 years old (Hicks, von Baeyer, Spafford, van Korlaar, & Goodenough, 2001; Tsze et al., 2013). It comprises a horizontal series of six-line drawings of faces representing increasing levels of pain, from "no hurt" to "very much hurt," scored 0–2–4–6–8–10, respectively. A difference of two points (or one face) is considered clinically important (von Baeyer, 2006; Hartling et al., 2013).

The secondary outcome, distress, was measured with the Observational Scale of Behavioral Distress-revised (OSBD-r) (Jay, Elliott, & Varni, 1986), which includes eight behaviors: information seeking, cry, scream, restraint, verbal resistance, seeking emotional support, verbal pain, and flail. A score of 0 indicates "no distress behavior" up to a maximum of 23.5. The OSBD-r has been validated for the assessment of procedural distress in the pediatric ER (Elliott, Jay, & Woody, 1987; Hartling et al., 2013; Noguchi, 2006). The OSBD-scores were obtained from video recordings for the following phases: (a) 2 min immediately before the procedure, (b) during the whole procedure, and (c) 2 min immediately after the procedure. The OSBD-r items are scored in 15 s intervals.

Heart rate (HR) was measured by the researcher conducting the trial before and after the procedure as a second indicator of distress, using a transcutaneous oximeter placed on the child's finger. Heart rate is considered a clinically relevant outcome measure for pain-related distress (Cohen et al., 2008; Cowen, Stasiowska, Laycock, & Bantel, 2015; Sweet & McGrath, 1998). Furthermore, we took clinical notes on whether a caretaker was present if any study-related adverse effects occurred and whether the attending nurse restrained the child during the procedure. Three researchers assessed the video footage for primary and secondary outcomes AHTPS and OSBD-r. By scoring the video material at a later time, we avoided having to assess observed pain and distress at the moment in a busy ER setting. The researchers all received training in scoring the AHTPS and OSBD-r.

Ethical Approval

The medical ethics committee of the University of Cape Town and the medical ethical committee of the RCWMCH approved the study in March 2014. Due to a low enrolment in the surgical trauma ward, we added the RCWMCH medical trauma ward as an additional site. We obtained the required ethical approval for this amendment on July 27, 2014. This change to the protocol did not affect any study methods and was made without breaking the blind on the accumulating data on participants' outcomes. The trial was registered in the Pan African Clinical Trials Registry (PACTR) under number PACTR201408000830410.

Sample Size and Power

Sample size calculations were based on the primary outcome pain, scored with the AHTPS (Stewart et al., 2004; Thiadens et al., 2011). A 30% reduction in pain score is considered clinically meaningful, which corresponds with a standardized mean difference at 0.54 (Cepeda & Carr, 2006). Under the assumption of a 30% reduction in pain score, a sample size of 55 per arm is needed to detect a significant difference in pain score with a power of 0.80 and a significance level at 0.05 (two-tailed). To allow for a 10% dropout, the study was designed for 60 patients per arm, thus a total of 180 patients.

Randomization and Allocation Concealment

Patients were randomized using consecutively numbered sealed opaque envelopes prepared by an independent researcher, who used a random number generation table for simple randomization. The envelopes were kept in a secure place, only accessible to the researchers (NvH and HM) who enrolled the patients.

Blinding of the observers to the nature of the intervention was not possible. To increase the reliability of the assessments, one researcher (M. van der Heijden) assessed all video recordings, while three other researchers each assessed footage of patients from only one of the study arms.

Statistical Methods

All data were analyzed according to the intention-to-treat principle. The focus in this trial was on comparisons between the two intervention groups and the control group, and not on differences in outcomes between before, during and after the procedure. Descriptive statistics are presented as *M* (*SD*) for normally distributed variables and as median [interquartile range (*IQR*)] or nonnormally distributed variables. The intraclass correlation (two-way mixed-effect model with an absolute agreement definition, reporting single measures) was calculated between the observers who assessed all video footage for the AHTPS and the OSBD-r.

The primary outcome pain AHTPS scores during and FPS-R after scores, and secondary outcome distress scores OSBD-r scores during and HR after the procedure were compared between study arms using Kruskal–Wallis tests, with post hoc comparisons between the two intervention groups and the control group using Dunn's test. Because relatively many children were restrained during the procedure, the ages of the restrained and nonrestrained children were compared using the Mann–Whitney *U* test. Multivariable linear regression analyses were performed with the outcome variables AHTPS during the procedure, OSBD-r during the procedure, FPS-R after the procedure, and HR after the procedure. In the regression analyses, we differentiated between a simple model (Model 1) and an extended model (Model 2). Model 1 included the independent variables study arm and, if applicable, the baseline values respectively OSBD-r before, FPS-R before, or HR before. Model 2 included the independent variables of Model 1 and type of procedure (IV placement; venepuncture; injection of local anesthetics POP back-slab; and wound dressing/suturing open wound), age, and sex. For the primary outcome, AHTPS Model 1 was not applicable because the residuals were nonnormally distributed. In Model 2, we included type of procedure because the procedures might vary in level of painfulness and duration. We also included age because it is known that younger children are often more distressed than older children. Plots of the model residuals of the linear regression models were checked for normality. A *p*-value of <.05 (two-tailed) was considered statistically significant, but we applied a Bonferroni correction to the significance level (adjusted significance level: $0.05/2 = 0.025$) for analyses with multiple comparisons of the two intervention groups to the control group. All data were analyzed in SPSS 23.0.

Results

Between April and September 2014, we included 197 children, 191 of whom received the intervention as intended and whose data were included in the analysis (see Figure 1). A total of 688 children were eligible for inclusion. The major reason for having missed eligible patients ($n = 491$) was admission during the weekends or nights when the researchers were not present. Only one family refused to participate and did not provide us a reason. None of the children asked to interrupt the intervention they were assigned to and were, therefore, all able to listen to the music or to watch the cartoon. In the cartoon group, we did not observe children who did not watch the cartoon.

The median age of the included 191 patients was 7.3 years (4.9–9.7); 68% were boys (see Table I). Of the five selected procedures, the POP back-slab was

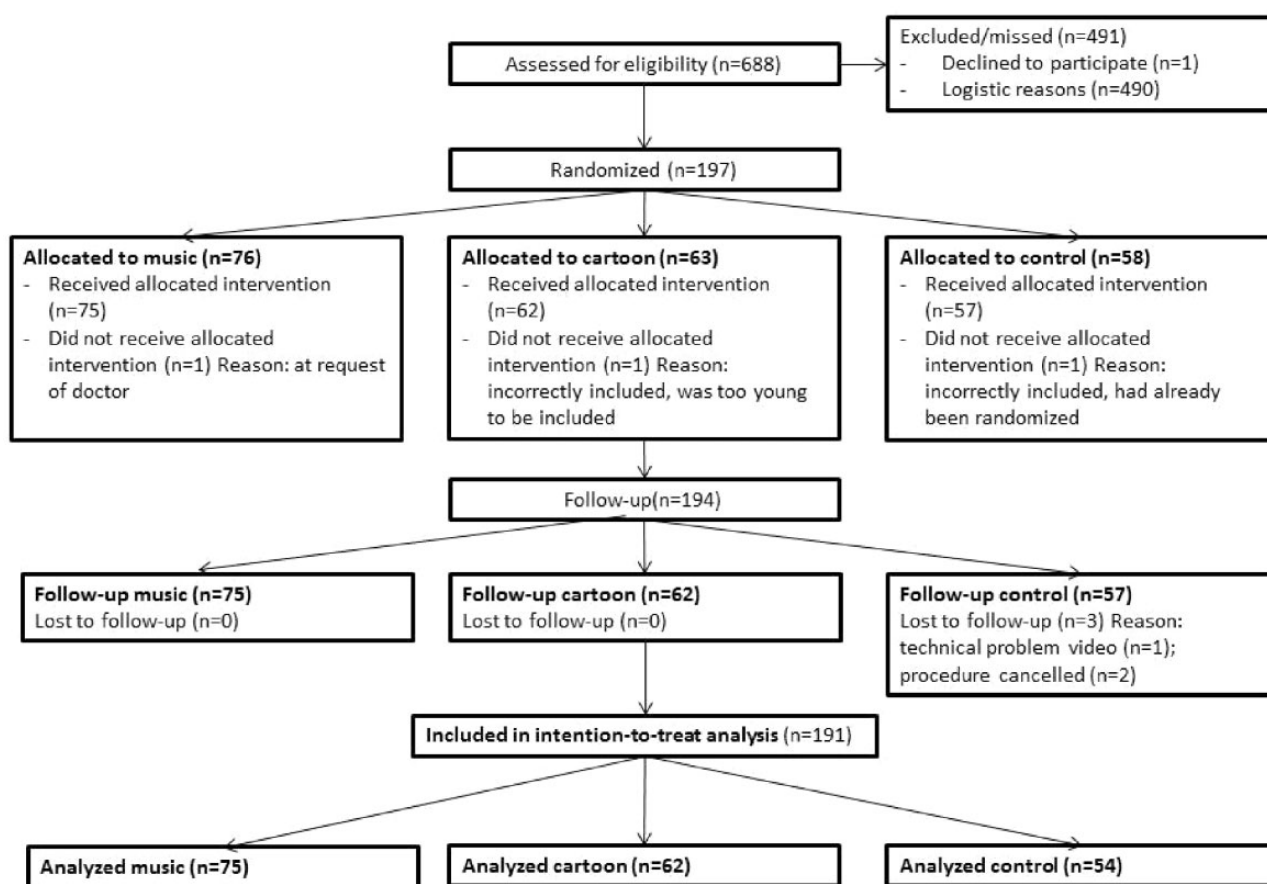


Figure 1. Flowchart.

most often performed (44%) and venepuncture the least often (10%). In 23.8% of patients, AHTPS pain scores were 0, indicating no pain during the procedure. Children older than 4 years self-reported FPS-R scores before ($n=117$) and after the procedure ($n=108$). In 39.3% of the patients before and 43.5% after the procedure, the FPS-R was scored 0.

Distress, as measured with the OSBD-r, was scored 0 (no distress) in 78% before the procedure, in 39% during and in 75% of the patients after the procedure. There was no significant difference in duration of the procedures between the three groups ($p=.082$) (see Table I).

One of the items of the OSBD-r concerns physical restraint (e.g., child being held down by parent or nurse). Thirty-three children (17%) were restrained during the procedure. These children were significantly younger ($p<.001$) than the children who were not restrained, median age 5.3 years [IQR 4.2–7.0] and 7.7 years (IQR 5.4–9.9), respectively. No study-related adverse effects occurred.

Pain and Distress Outcomes

Three observers separately assessed the pain outcome AHTPS and distress outcome OSBD-r from video footage. The interrater reliability was good with intra-class correlations of 0.94 (AHTPS), 0.76 (OSBD-r

before), 0.90 (OSBD-r during), and 0.89 (OSBD-r after). Data for six patients were missing for the primary outcome AHTPS.

Primary Outcome: Pain

The AHTPS scores during the procedure were significantly different between the groups (Kruskal–Wallis $p=.017$); pairwise comparisons showed that patients in the music group expressed significantly less pain during the procedure than the control group ($p=.006$) (see also Figure 2 and Table II). Multivariable linear regression analysis Model 2 with AHTPS during procedure as outcome variable also revealed statistically significantly less pain ($B=-1.173$, 95% confidence interval $-1.953, -0.394$, $p=.003$) for the music group as compared to the control group after adjusting for type of procedure, age and sex (Table III). In Model 2, the lower pain score of the cartoon group ($B=-0.377$) compared to the control group was not statistically significant ($p=.353$).

Secondary Outcomes: Pain and Distress

Pain scores measured by the FPS-R and distress scores measured by the OSBD-r, and HR before, during and after the procedure did not significantly differ between the three intervention groups (see Table II). For FPS-

Table I. Patient Characteristics and Clinical Characteristics

	Control (n = 54)	Cartoon (n = 62)	Music (n = 75)	Total (n = 191)
Sex				
Boy (%)	40 (74)	42 (68)	48 (64)	130 (68)
Age (years), median (IQR)	7.7 (5.1–9.8)	6.6 (4.8–8.6)	7.5 (5.0–10.1)	7.3 (4.9–9.7)
First language, n (%)	n = 51	n = 59	n = 71	n = 181
English	17 (33)	26 (44)	26 (37)	69 (38)
Xhosa	15 (29)	18 (31)	28 (39)	61 (34)
Afrikaans	18 (35)	15 (25)	16 (23)	49 (27)
Other	1 (2)	–	1 (1)	2 (1)
Type of procedure, n (%)				
POP back-slab	27 (50)	28 (45)	29 (39)	84 (44)
IV placement	6 (11)	10 (16)	21 (28)	37 (19)
Wound dressing, suturing open wounds	11 (21)	12 (20)	7 (9)	30 (16)
Injection of local anesthetics	5 (9)	5 (8)	11 (15)	21 (11)
Venipuncture	5 (9)	7 (11)	7 (9)	19 (10)
Reason for admission, n (%)				
Arm fracture	24 (44)	25 (40)	29 (39)	78 (41)
MVA	7 (13)	10 (16)	5 (7)	22 (12)
Leg fracture	4 (7)	5 (8)	7 (9)	16 (8)
Laceration	2 (4)	2 (3)	6 (8)	10 (5)
Burn	4 (7)	3 (5)	2 (3)	9 (5)
Head injury	1 (2)	4 (7)	4 (5)	9 (5)
NAI	2 (4)	2 (3)	4 (5)	8 (4)
Dog bite	2 (4)	1 (2)	3 (4)	6 (3)
Other	8 (15)	10 (16)	15 (20)	33 (17)
Parental presence, n (%)	44 (82)	59 (95)	67 (89)	170 (89)
Restraint, n (%)	8 (15)	10 (16)	15 (20)	33 (17)
Duration of procedure (min), median (IQR)	05:12 (02:58–08:23)	05:27 (02:52–09:01)	03:43 (01:43–07:11)	04:22 (02:08–07:51)

Note. IQR = interquartile range; MVA = motor vehicle accident, NAI = nonaccidental injury.

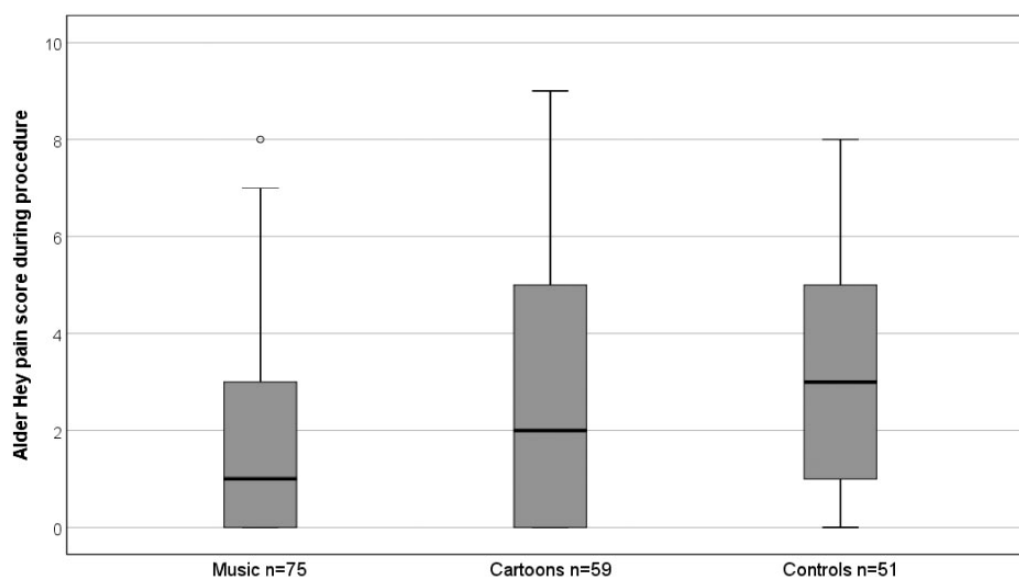


Figure 2. Boxplot showing the pain scores measured with the AHTPS during the procedures. The music group shows statistically significantly less pain. AHTPS = Alder Hey Triage Pain Score.

R, post hoc comparisons based on Dunn's test did not show a statistically significant difference between the music group and the control group ($p = .03$), and between the cartoon group and the control group ($p = .50$) with an adjusted significance level set at 0.025 (Table II).

Regression analysis Model 1 with FPS-R scores after the procedure as outcome did not show an overall statistically significant difference between the intervention groups ($p = .064$) (Table IV). In Model 2, the difference between the music group and the control group was not statistically significant ($p = .025$) after

Table II. Outcomes Per Intervention Group

Outcomes	Control (<i>n</i> = 54)	Cartoon (<i>n</i> = 62)	Music (<i>n</i> = 75)	Total (<i>n</i> = 191) ^a	<i>p</i> -value ^b
AHTPS during (<i>n</i>)	51	59	75	185	
<i>M</i> (<i>SD</i>)	3.10 (2.24)	2.86 (2.62)	2 (2.05)		
Median (IQR)	3 (1–5)	2 (0–5)	1 (0–3)	2 (1–4)	
Range	3.10 (0–8)	2.86 (0–9)	2 (0–8)	2.58 (0–9)	.017 ^c
OSBD-r before (<i>n</i>)	49	56	74	179	
<i>M</i> (<i>SD</i>)	0.27 (0.70)	0.40 (1.05)	0.36 (0.85)		
Median (IQR)	0 (0–0)	0 (0–0)	0 (0–0.3)	0 (0–0)	.985
Range	0–2.70	0–5.00	0–4.94	0–5	
OSBD-r during (<i>n</i>)	51	57	74	182	
<i>M</i> (<i>SD</i>)	1.34 (1.89)	1.63 (2.51)	1.24 (1.67)		
Median (IQR)	0.37 (0–2.63)	0.33 (0–2.54)	0.49 (0–2.51)	0.38 (0–2.54)	.550
Range	0–7.33	0–10.50	0–9.12	0–10.50	
OSBD-r after (<i>n</i>)	48	56	74	178	
<i>M</i> (<i>SD</i>)	0.39 (0.85)	0.77 (1.70)	0.41 (1.16)		
Median (IQR)	0 (0–0)	0 (0–0.74)	0 (0–0.05)	0 (0–0.05)	.749
Range	0–3.00	0–8.75	0–7.40	0–8.75	
FPS-r before (<i>n</i>)	33	39	45	117	
<i>M</i> (<i>SD</i>)	2.94 (3.39)	3.90 (3.97)	2.62 (3.21)		
Median (IQR)	2 (0–4)	2 (0–8)	2 (0–4)	2 (0–5)	.462
Range	0–10	0–10	0–10	0–10	
FPS-R after (<i>n</i>)	30	35	43	108	
<i>M</i> (<i>SD</i>)	3.93 (3.77)	3.54 (3.79)	2.19 (3.23)		
Median (IQR)	3 (0–6)	2 (0–6)	0 (0–2)	2 (0–6)	.077 ^d
Range	0–10	0–10	0–10	0–10	
HR before (<i>n</i>)	53	57	70	180	
<i>M</i> (<i>SD</i>)	102.43 (21.11)	106.77 (21.47)	105.81 (20.62)		
Median (IQR)	100 (84–115)	106 (93–118)	103 (93–118)	102 (91–116)	.450
Range	66–158	62–161	66–164	62–164	
HR after (<i>n</i>)	53	56	69	178	
<i>M</i> (<i>SD</i>)	112.87 (23.90)	114.20 (27.50)	110.07 (21.52)		
Median (IQR)	112 (94–129)	109 (94–137)	111 (94–125)	111 (94–130)	.825
Range	66–172	65–169	62–164	62–172	

Note. AHTPS = Alder Hey Triage Pain Score; FPS-R = Faces Pain Scale-Revised; HR = heart rate; IQR = interquartile range; *SD* = standard deviation; OSBD-r = Observational Scale of Behavioral Distress-revised.

^aAHTPS scores were missing for six patients in the intention-to-treat analysis.

^bKruskal–Wallis test for any difference between the three groups.

^cPost hoc pairwise comparisons using the Dunn's test showed a significant difference between the music group and the control group ($p = .006$) and no significant difference between the cartoon group and the control group ($p = .328$), with an adjusted significance level of 0.025.

^dPost hoc pairwise comparisons using the Dunn's test showed no significant difference between the music group and the control group ($p = .03$) and between the cartoon group and the control group ($p = .50$), with an adjusted significance level of 0.025.

adjustment for multiple testing. The difference in FPS-R scores between the cartoon group and control group was not statistically significant in both models ($p = .518$ and $p = .51$).

For the secondary outcome distress, regression analysis with OSBD-r as outcome revealed no statistically significant difference in distress scores between the three trial arms (Table IV). In addition, for the outcome HR, regression analysis with HR after the procedure as outcome revealed no statistically significant difference between the study arms before and after procedures.

Independent Variables

Higher FPS-R, OSBD-r, and HR scores before the procedure were significantly and positively associated

with higher scores during the procedure in both models (Table IV). Younger children had significantly higher OSBD-r scores and HR than had older children ($p < .001$ and $p = .001$) (Table IV).

Type of procedure had a significant effect on AHTPS scores and OSBD-r scores in that putting on a cast was significantly less painful ($p = .004$) and less distressing ($p = .01$) than was wound dressing/suturing (Tables III and IV). Pain scores were not significantly different for boys and girls.

Discussion

In this RCT, we found that children receiving a music intervention during a procedure in the ER showed significantly lower observed pain on the AHTPS than

Table III. Multivariable Linear Regression Analysis Model 2 for Primary Outcome Pain: AHTPS During Procedure

Pain AHTPS during procedure			
	B	95% CI	p-value
Study arm			
Music	−1.173	−1.953, −0.394	.003
Cartoon watching	−0.377	−1.177, 0.422	.353
Control	Reference		
Type of procedure			
IV placement	0.076	−0.991, 1.143	.888
Venepuncture	−0.877	−2.141, 0.387	.173
Injection local anesthetics	0.101	−1.136, 1.337	.872
POP back-slab	−1.337	−2.251, −0.422	.004
Wound dressing/suturing	Reference		
Age (years)	−0.253		
Sex			
Boys	0.107		
Girls	Reference		

Note. AHTPS = Alder Hey Triage Pain Score; CI = confidence interval; OSBD-r = Observational Scale of Behavioral Distress-revised; POP = Plaster of Paris.

children receiving only standard care. This was not the case for self-reported pain on the FPS-R. Observed distress scores OSBD-r and HR did not significantly differ between the intervention and control groups. There was no significant effect on observed pain (AHTPS) or distress (OSBD-r) for children watching cartoons.

Almost half of the children did not display pain prior to the procedure and three-quarters of the children showed no signs of distress, therefore, there was little room for improvement to begin with. One of the explanations for this can be that the procedures were not equally distressing and that most children underwent the POP back-slab, the least painful and least distressing procedure. Notwithstanding, the ER procedures remained distressing and painful for a large group of children. A substantial number of children (17%), equally divided across the groups, were restrained for fear of movement during procedures. Although this does not directly affect the outcomes of our study, it is not a desirable approach because it is a source of distress.

The results of our study contrast the findings of some other studies. Downey et al. found that cartoon watching had a significant pain-reducing effect in young children in the ER, and Cohen et al. found this to be a beneficial distraction during immunizations (Cohen et al., 1997; Downey & Zun, 2012). Hartling et al. (2013) found music interventions to have a significant distress-reducing effect during IV placement in children aged 3–11 years in the ER, but only after patients with low OSBD-r scores had been excluded from the analysis. We refrained from this type of

analysis because this would substantially reduce the numbers of patients and consequently result in a lack of power. In our study, the control group had lower median OSBD-r scores during the procedure; 0.37 (IQR 0–2.63) compared to Hartling's median of 2.21 (IQR 0.18–3.83). This is probably due to inclusion of less distressing procedures such as putting on an orthopedic cast in our study. We chose not to limit the interventions to IV placement because we wanted to be able to generalize our findings.

Notwithstanding extensive research, we still have limited understanding of what type of distraction works and for whom in the pediatric ER when, how and for whom distraction works in the pediatric ER (Birnie et al., 2017). Also, barriers and successes for implementation of distraction interventions are scarcely reported (Bergomi et al., 2018; Birnie et al., 2018). There is potential for successful implementation of distraction interventions such as listening to recorded music because it requires minimal resources and is widely available (Birnie et al., 2018). However, using headphones to listen to music restricts communication between the doctor and the patient and playing music over speakers might interfere with the staff's concentration and communication. A barrier for visual distraction might be that the child is unable to see the intervention. In our study, all children were properly assigned to their interventions. For those assigned to watch cartoons, the researcher made sure these were clearly visible. Although we did not observe children ignoring the cartoons, it is difficult to ensure they paid full attention.

Despite using a combination of four widely accepted outcome measurements to measure pain and distress, we were only able to show a significant effect in one pain outcome measurement. The question remains which outcome measurement is the best reflection and when to measure this difference. Pain and distress are concepts that are intertwined and difficult to distinguish in how they are expressed, especially in young children. This is also apparent in the outcome measurements. Despite an overlap in the items of observational pain and distress scales, the AHTPS is validated for pain during painful procedures in the ER, whereas the OSBD-r is validated to measure procedure-related distress. However, there is a difference in scoring procedures. The AHTPS scores five items as being present or absent at any time during the observation period. The AHTPS scores the most painful behavior but does not specify the duration of the observation period. In contrast, the OSBD-r scores eight items per every 15 s of the entire procedure, taking into account the duration of the behavior. Also, the OSBD-r items are weighted according to intensity. In other words, the OSBD-r is an expression of an average over time, whereas the AHTPS shows the maximum score at one point in time. Heart rate is widely

Table IV. Regression Analyses Model 1 and Model 2 for Pain Score Outcome FPS-R and Distress Score Outcomes OSBD-r and HR

	Model 1			Model 2		
	B	95% CI	p-value	B	95% CI	p-value
Pain FPS-R after procedure						
Study arm						
Music	−1.537	−3.141, 0.067	.064	−1.96	−3.67, −0.25	.025
Cartoon watching	−0.548	−2.224, 1.127	.518	−0.58	−2.31, 1.15	.51
Control	Reference			Reference		
Type of procedure						
IV placement				1.94	−0.46, 4.33	.12
Venepuncture				1.27	−2.10, 4.64	.46
Injection local anesthetics				2.27	−0.64, 5.17	.13
POP back-slab				0.98	−1.10, 3.06	.35
Wound dressing/suturing				Reference		
FPS-R before	0.338	0.148, 0.528	.001	0.32	0.12, 0.51	.002
Age (years)				0.01	−0.30, 0.32	.96
Sex						
Boys				−0.33	−1.75, 1.09	.65
Girls				Reference		
Distress OSBD-r during procedure						
Study arm						
Music	−0.137	−0.773, 0.499	.671	−0.23	−0.84, 0.38	.45
Cartoon watching	0.092	−0.584, 0.768	.789	0.031	−0.60, 0.66	.93
Control	Reference			Reference		
Type of procedure						
IV placement				0.19	−0.63, 1.01	.65
Venepuncture				−0.16	−1.22, 0.89	.76
Injection local anesthetics				0.68	−0.29, 1.64	.17
POP back-slab				−0.92	−1.63, −0.22	.01
Wound dressing/suturing				Reference		
OSBD-r before procedure	1.010	0.715, 1.306	<.001	0.82	0.54, 1.11	<.001
Age (years)				−0.16	−0.25, −0.073	<.001
Sex						
Boys				−0.08	−0.60, 0.45	.77
Girls				Reference		
Distress HR after procedure						
Study arm						
Music	−5.431	−11.907, 1.044	.100	−5.54	−11.89, 0.82	.09
Cartoon watching	−3.502	−10.344, 3.339	.314	−3.47	−10.0, 3.08	.30
Control	Reference			Reference		
Type of procedure						
IV placement				3.89	−4.69, 12.47	.37
Venepuncture				−5.47	−16.31, 5.38	.32
Injection local anesthetics				3.18	−7.33, 13.69	.55
POP back-slab				−7.01	−14.24, 0.22	.06
Wound dressing/suturing				Reference		
HR before procedure	0.771	0.642, 0.900	<.001	0.66	−0.52, 0.80	<.001
Age (years)				−1.76	−2.74, −0.78	.001
Sex						
Boys				−1.72	−7.48, 4.04	.56
Girls				Reference		

Note. Model 1 was adjusted for intervention group and baseline value. Model 2 was adjusted for Model 1, type of procedure, age, and sex. FPS-R = Faces Pain Scale-Revised; HR = heart rate; OSBD-r = Observational Scale of Behavioral Distress-revised; POP = Plaster of Paris.

used and appreciated as a clinical indicator for distress and pain (Cohen et al., 2008). However, there is also discussion on the use of vital signs as a predictor for distress and pain, especially in an acute setting where pain and distress strongly influence each other (Bartfield, Janikas, & Lee, 2003; Bossart, Fosnocht, & Swanson, 2007). The FPS-R is recommended to use as a self-report pain intensity measurement, especially in

younger children (Hicks et al., 2001). However, the Visual Analog Scale (VAS) and Color Analog Scale (CAS) are deemed more age-appropriate and reliable for children older than 8 years of age (Le May et al., 2018). Self-report is the gold standard of pain assessment. However, in children 3–7 years of age von Baeyer, Uman, Chambers, & Gouthro, (2011) concluded that “no validated screening tasks exist to

distinguish children who can accurately use self-report pain measurements from those who cannot.” Furthermore, although the ability to use self-report develops from age 3–7 years, it has also been reported that the reliability and validity of self-report scales are overestimated in children younger than 7 years old (Tsze et al., 2013; von Baeyer et al., 2017).

Limitations

This study was bound by some limitations. Firstly, it was not possible to blind the patients, their parents, and the medical staff to the interventions, nor could the observers who assessed the videos be blinded. This limitation is inherent to the type of intervention. Secondly, filming in itself may have influenced the behavior of patients, parents, and medical staff. One might be more conscious of how one is perceived knowing your behavior is filmed, albeit for research purposes. Thirdly, it is possible that some children could have been too old to appreciate the cartoons or that it might have been better to use self-selected music and perhaps headphones as well.

Conclusions and Recommendations

Distraction interventions such as recorded music could be beneficial for children experiencing pain and pain-related distress during ER procedures. Future studies should investigate if using self-selected music or parent-selected music might increase the effectiveness. Recorded music is noninvasive, can be administered at low costs and can be implemented independently by nurses, which is particularly useful in resource-limited settings.

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